TP1: Virtual Machine Deployment, Networking, and SSH Configuration Using VMware

1. Objectives

By the end of this practical lab, you should be able to:

- Install and configure a Type 2 hypervisor.
- Create a virtual machine and install a Linux operating system.
- Allocate and manage virtual CPU, memory, and storage resources.
- Understand and configure NAT, Bridged, and host-only/custom networking.
- Install and test an SSH server.
- Export, import, and configure a second virtual machine.
- Establish network connectivity between the host and virtual machines.
- Create and restore a snapshot to recover a known working state.

2. Lab scenario

You are a junior DevOps engineer preparing a small virtualized Linux environment for testing and remote administration. Your task is to deploy two Linux virtual machines, configure their network interfaces, establish secure remote connections, and test how snapshots can be used to recover a virtual machine after a configuration change.

The final environment will contain:

- **LinuxLite-01:** the primary virtual machine, accessible from the host.
- **LinuxLite-02:** a second virtual machine created by exporting and importing the first VM.
- **Host machine:** the physical computer running the hypervisor.

Part A: Install the hypervisor and prepare the environment

Step 1. Verify hardware virtualization

1. Check that hardware virtualization is enabled in your computer's BIOS/UEFI.
2. On Windows, open Task Manager → Performance → CPU and look for the virtualization status.
3. Download and install VMware Workstation or VirtualBox.
4. Launch the hypervisor and verify that it works correctly.

Q1. How did you verify that hardware virtualization was enabled? What is the role of a hypervisor in this lab?

Answer:
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Step 2. Create the first virtual machine

1. Download the Linux Lite ISO image from its official website.
2. Create a new virtual machine named LinuxLite-01.
3. Configure the following resources:

Resource	Configuration
Virtual CPUs	2 vCPUs
RAM	2 GB
Virtual disk	20 GB, dynamically allocated if supported
Guest OS	Linux Lite
Network adapter	NAT initially

4. Attach the ISO image to the virtual CD/DVD drive.
5. Start the VM and complete the Linux installation.
6. Use your family name as the Linux username.
7. Set the hostname to LinuxLite-01. If necessary, run: sudo hostnamectl set-hostname LinuxLite-01

8. Restart the VM if required and verify the hostname: hostname

9. Measure the approximate time between starting the VM and reaching the login prompt or usable desktop.

Q2. Record the resources allocated to your VM. Why does a virtual machine need allocated CPU, memory, and disk space?

Answer:
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.....

Boot time: ..... seconds

Part B: Configure networking and SSH

Step 3. Configure NAT and inspect network settings

1. Keep the first network adapter of LinuxLite-01 in NAT mode.
2. Start the VM and open a terminal.
3. Display its network interfaces and IP addresses: ip a

4. Test connectivity to an external IP address, if Internet access is available: ping -c 4 8.8.8.8

5. Observe the IP address assigned to the VM.

Q3. What IP address did the VM obtain? Explain how NAT allows a VM to access an external network.

Answer:
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Step 4. Install and test the SSH server

Inside LinuxLite-01, execute:

sudo apt update

sudo apt install -y openssh-server

Check the SSH service:

sudo systemctl status ssh

If necessary, start and enable it:

sudo systemctl enable --now ssh

Test SSH locally:

ssh localhost

Accept the host key confirmation if prompted, then enter your Linux account password. After confirming the connection, exit the SSH session: exit

Q4. What output indicates that the SSH service is running? What happens when you execute ssh localhost?

Answer:
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Part C: Create a second virtual machine

Step 5. Export and import the first VM

1. Shut down LinuxLite-01 properly.
2. Export the VM as an OVA appliance if your hypervisor supports this format.
3. Import the appliance to create a second virtual machine.
4. Name the imported VM LinuxLite-02.
5. If prompted, choose the option to generate new virtual machine identifiers or MAC addresses so the two VMs can coexist.
6. Start LinuxLite-02 and verify that the operating system works.
7. Change its internal hostname:

sudo hostnamectl set-hostname LinuxLite-02

8. Verify the hostname: hostname

9. Shut down and restart the VM if required.

Q5. What is the difference between the VM's name in the hypervisor and its hostname inside Linux? Why should the two VMs have different hostnames?

Answer:
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Part D: Configure communication between the machines

Step 6. Configure the network adapters

Shut down both virtual machines before changing their network settings.
Configure the network adapters as follows:

Virtual machine	Adapter 1	Adapter 2
LinuxLite-01	Custom / host-only network shared with LinuxLite-02	Bridged
LinuxLite-02	The same Custom / host-only network	Not required

The purpose of this configuration is to create a private network for communication between the two VMs while providing LinuxLite-01 with a separate interface connected to the physical local network.

Make sure the custom network is configured to allow communication between the VMs. Enable host access to this network if your hypervisor supports it.

Start both VMs and run the following command inside each: ip a

Record the relevant interface names and IP addresses.

Machine	Interface	IP address
LinuxLite-01	Custom / host-only	.....
LinuxLite-01	Bridged	.....
LinuxLite-02	Custom / host-only	.....

Q6. Explain the difference between NAT, Bridged, and host-only networking. Which network is used for communication between the two VMs?

Answer:
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Step 7. Test connectivity from the host

From the host operating system, ping the **Bridged IP address** of LinuxLite-01:

ping <LINUXLITE01_BRIDGED_IP>

Replace the placeholder with the actual IP address.

Record the command and output. If the ping fails, check the network configuration and firewall rules before proceeding.

Q7. What command did you use, and was the VM reachable from the host?

Answer:
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Step 8. Connect to LinuxLite-01 using SSH

From a terminal on the host machine, execute:

ssh <username>@<LINUXLITE01_BRIDGED_IP>

Replace the placeholders with your Linux username and the VM's Bridged IP address.

After connecting, execute:

hostname
whoami
ip a

Verify that the hostname is LinuxLite-01 and that the displayed user is your Linux account. Then disconnect:

exit

Q8. How can you confirm that you are logged into LinuxLite-01 rather than the host machine?

Answer:
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Step 9. Connect from LinuxLite-01 to LinuxLite-02

1. Ensure that the SSH server is installed and running on LinuxLite-02, using the commands from Step 4.
2. From LinuxLite-01, connect to LinuxLite-02 using its Custom / host-only IP address:

ssh <username>@<LINUXLITE02_CUSTOM_IP>

3. After connecting, execute: hostname Then: whoami

4. Confirm that the hostname is LinuxLite-02.
5. Disconnect using exit.

Q9. Record the SSH command and the output confirming that the connection reached LinuxLite-02.

Answer:
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Part E: Snapshots and recovery

Step 10. Create a baseline snapshot

1. Choose LinuxLite-01 as the VM to use for this experiment.
2. Ensure that the VM is in a clean, working state.
3. Take a snapshot using your hypervisor's snapshot feature.
4. Name the snapshot baseline.

A snapshot records a particular state of a VM so that you can return to it later. It is useful for testing, but it is not a substitute for a complete backup.

Q10. Why is it useful to take a baseline snapshot before making changes to a virtual machine?

Answer:
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Step 11. Make a change and restore the snapshot
Inside LinuxLite-01, install the Apache web server:

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sudo apt update
sudo apt install -y apache2
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Check its status:

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sudo systemctl status apache2
```

Stop the service to simulate a configuration or service failure:

```
sudo systemctl stop apache2
sudo systemctl status apache2
```

Observe the service state. Next:

1. Shut down the VM if required by your hypervisor.
2. Restore the snapshot named baseline.
3. Start the VM and check the Apache service again:

```
sudo systemctl status apache2
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If Apache was not installed when the snapshot was created, it should not be present after restoring that snapshot. Confirm your result rather than assuming it.

Q11. What changed after the snapshot was restored? Explain how snapshots can help DevOps engineers test changes safely.

Answer:
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